Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	*(a)*			Ticks in boxes 2, 3, 4 (-1 for each additional box ticked) (3) The largest supergiant is Betelgeuse (✓) The radius of a white dwarf is approximately 100 times smaller than the radius of the Sun (✓) The hottest star is β Centauri (✓)		3		3		
	(b)	(i)		Currently the temperature is: stated between 5500 $\rightarrow$ 6 000 K (1) When it becomes a [red] giant its temperature will decrease to: stated between 3 000 $\rightarrow$ 5 000 K (1) When it becomes a white dwarf its temperature will increase to: stated between 6 100 $\rightarrow$ 30 000 K, so disagree (1) Award for 1 mark for stating when it becomes a red giant, temperature decreases and when it becomes a white dwarf, temperature increases			3	3	3	
		(ii)		Radiation / pressure becomes greater than the gravitational [force] (1) So it will <u>expand</u> into a <u>red</u> giant (1) In the final stage the radiation / pressure becomes smaller [than the gravitational force] so it contracts into a white dwarf (1)	3			3		
	(c)			During the <u>supernova</u> stage in the life cycle of a <u>large</u> star (1) material [including heavy elements] is returned into space (1) The solar system originated from the <u>collapse</u> of a cloud of gas and dust from such an event (1)	3			3		
				Question 5 total	**6**	**3**	**3**	**12**	**3**	**0**